DEPI | AI In Education

DEPI GRADUATION PROJECT

Task 2 - Literature Review

Automated Teacher Workflow with n8n

1. Feedback & Evaluation

Project context. Task 2 continues directly from the Project Planning & Management document submitted in Task 1. The same project scope, technical direction, team structure, risks, KPIs, and project boundaries are retained unless the lecturer requests a change.

1.1 Lecturer Feedback

Feedback received from the lecturer (Eng. Alaa)

The lecturer stated that he has no comments on Task 1. Accordingly, no revisions to the submitted Task 1 are required at this stage, and the team will continue with the same project direction and scope.

1.2 Evaluation Summary

Evaluation Area	Current Status	Team Action
Project idea and direction	No lecturer comments on Task 1	Continue with Project #9 as submitted.
Project scope	No revision requested	Retain the core n8n workflow scope and outputs defined in Task 1.
Technical approach	No revision requested	Proceed with trigger, connected nodes, AI transformations, branching, conditions, error handling, fallback path, and formatted teacher review output.
Planning assumptions	No revision requested	Use the existing timeline, risks, KPIs, GitHub approach, and project boundaries as the implementation baseline.

Task 2 interpretation

The lecturer feedback is recorded exactly as received: no comments on Task 1. The improvement items in the next section are therefore team-proposed enhancements for implementation quality, not lecturer-requested corrections.

2. Suggested Improvements

Because no corrective feedback was requested for Task 1, the following items are implementation-focused enhancements that strengthen reliability, consistency, and teacher usability without changing the approved project direction.

ID	Suggested Improvement	Why It Improves the Project	Priority	Implementation Stage
I1	Strengthen input validation and clarification paths	Prevents incomplete or unclear teacher inputs from reaching the AI generation steps.	High	Before core AI generation
I2	Use structured and reusable prompt templates	Keeps the five AI stages consistent and makes prompt refinement easier to document and test.	High	Prompt design and implementation
I3	Standardize data passed between workflow nodes	Reduces formatting errors and keeps subject, grade, topic, objective, and lesson context consistent across outputs.	High	Workflow design and build
I4	Define explicit branching rules for differentiation	Ensures the differentiated version follows clear conditions rather than an uncontrolled free-form generation step.	High	Branching and conditions
I5	Expand error handling, retry, and fallback behavior	Helps the workflow recover from model/API or node failures without returning a misleading final package.	High	Robustness and QA
I6	Create a consistent teacher-review output template	Makes the generated lesson package easier to scan, review, edit, and approve before classroom use.	Medium	Output formatting
I7	Use a broader test matrix before final delivery	Validates normal, edge, invalid-input, branch, and failure scenarios across more than one lesson context.	Medium	Testing and refinement
I8	Keep external platform/API integration optional	Protects the core deadline while leaving a clear enhancement path after the main workflow is stable.	Optional	After core workflow completion

3. Final Grading Criteria

Important note on grading weights

The DEPI documentation guidelines identify the final grading areas as Documentation, Implementation, Testing, and Presentation. The provided guideline does not state percentage weights, so this Task 2 does not invent or assign percentages.

Grading Area	What the Project Should Demonstrate	Evidence to Prepare	Current Preparation Status
Documentation	Clear and consistent project documentation that reflects the actual workflow and implementation decisions.	Task 1 planning file; Task 2; Task 3 requirements; workflow diagram; prompts; README/user guidance; GitHub documentation.	In progress
Implementation	A working n8n automation with trigger, multiple nodes, data passing, AI transformations, branching/conditions, error handling, fallback path, and organized outputs.	Working workflow; exported n8n JSON; prompt versions; sample generated lesson package.	Planned for implementation phase
Testing	Evidence that the workflow was tested under normal, edge, invalid-input, branch, and failure scenarios, with issues corrected.	Test cases and results; error/fallback demonstration; issue and fix log; final regression run.	Planned for testing phase
Presentation	A clear explanation and demonstration of the problem, solution, workflow logic, implementation choices, challenges, and project outcomes.	Final PPT/PDF; live or recorded demo; concise explanation of team roles and project results.	Planned for final delivery

3.1 Task 2 Outcome

- Lecturer feedback has been documented: no comments were provided on Task 1.
- No changes are required to the Task 1 project direction at this stage.
- The team will use the suggested improvements above as implementation-quality enhancements, not as changes to the approved scope.
- The project will continue to Requirements Gathering and System Analysis & Design using the Task 1 baseline.